

Same Budget, More Water

Turning WaPOR's evaporation-transpiration split into a ranked spending list for Egypt's irrigation modernisation budget

Field	Entry
Subtheme (primary)	Governance
Subtheme (secondary)	Dashboarding & Entrepreneurship; Communication (cross-cutting)
Governance category	Category 2 — National: ministries and government agencies
Policy-cycle phase	Implement, then Monitor
Tool direction	Accountability — performance assessment and KPIs
Tool selected	IPAT — Irrigation assessment tool for Egypt (MWRI, WaPOR v3 L3 20 m)
Case study	Nile Delta and Valley, Egypt
Target actor	Planning officer, Ministry of Water Resources and Irrigation
WaPOR layers used	AETI, E, T, I — Level 2, 100 m, dekadal
SDG linkage	6.4.1 water-use efficiency; 2.4 sustainable agriculture

1. The core problem

The tools already exist. FAO's catalogue lists more than twenty working WaPOR applications, and Egypt has one of its own: IPAT, the irrigation assessment tool hosted by the Ministry of Water Resources and Irrigation, built on WaPOR v3 Level 3 at 20 metres across roughly 1.5 million hectares of the Middle and West Delta. It shows, accurately, where water productivity is high and where it is low.

It stops there. A planning officer with a fixed annual modernisation budget still has to decide where to spend it, and a productivity map does not answer that question. Today the budget is split by cultivated area or by political balance across governorates. Nothing in the toolchain says which hectare of subsidy buys the most water back.

The gap is not data, and it is not accuracy. It is that the output arrives in the wrong units for the decision — millimetres per dekad, when the decision is denominated in pounds, hectares and cubic metres.

2. The insight

WaPOR is one of very few global products that reports evaporation (E), transpiration (T) and interception (I) as separate layers. Almost nobody uses that split, and it is the most decision-relevant thing in the database.

Transpiration is water that passed through a stoma and grew a crop. Evaporation is water that was delivered, paid for, and left the field having grown nothing. Unlike most losses in an irrigation system, a large share of it is recoverable with cheap, well-understood measures: alternate wetting and drying in paddy rice, laser land levelling, mulching, drip retrofit.

So the question “where should the money go?” has a defensible, data-driven answer: wherever a unit of spending recovers the most evaporated water. That is a number WaPOR can produce today, everywhere, without new instrumentation.

3. The user and the decision

Element	Detail
User	Planning officer, Ministry of Water Resources and Irrigation
Decision-maker	The annual on-farm modernisation budget committee
Constraints	Fixed annual envelope; pressure to spread evenly across governorates; must justify spending to the Ministry of Finance; operates on an annual cycle, not a dekadal one
Decision informed	Which governorates receive this season’s modernisation budget, in what proportion, and with which measure
Moment	Annual budget planning — an existing process, on an existing calendar, requiring no new workflow

The test: *because of this, an MWRI planning officer will direct this season’s budget to Upper Egypt first — Aswan, Qena, Sohag, Assiut — instead of splitting it in proportion to cultivated area across every governorate.*

4. Where this sits in the governance framework

The subtheme asks for a real governance problem, a specific target actor, and a decision the idea helps that actor formulate, implement or monitor. This concept answers all three at Category 2, the national level.

Governance function	What this concept does
Formulate	Not addressed. The policy already exists — Egypt already subsidises on-farm modernisation.
Implement	The core contribution. “Direct financial and technical resources with precision by identifying hot spots of acute water stress” is precisely the ranked spending list. The evaporation hot spot is a spending instruction.
Monitor	WaPOR observes the same district next season. The indicator that justified the spending is the indicator that measures it, so the accountability loop closes on a single, verifiable biophysical source rather than on self-reported progress.

What cannot be captured in pixels. The subtheme is explicit that stakeholder engagement happens outside the dataset interface, and we take that seriously. A ranked list that ignores political reality will be ignored in turn. That is why the budget is a slider and the cost assumptions are editable inputs: the committee argues with the model in the room, changes what it disagrees with, and leaves owning the result. The tool is built to lose an argument gracefully — which is what makes it likely to be used at all.

5. Method

For every combination of district and applicable intervention:

$$\text{recoverable water } V = E_{\text{mm}} \times \text{irrigated_ha} \times 10 \times \text{reduction_fraction} \quad [\text{m}^3]$$

$$\text{annualised cost } C = (\text{USD/ha} \div \text{life_years}) \times \text{irrigated_ha} \quad [\text{USD/year}]$$

$$\text{cost of water } c = C \div V \quad [\text{USD/m}^3]$$

Sort every pair by c ascending and spend the budget down that list — cheapest water first. Because a district can be part-treated, the greedy solution is optimal. The baseline it is measured against splits the same budget in proportion to irrigated area. That is a neutral prior — what you would do knowing only where the cropland is — and not a claim about how the ministry currently allocates, for which we have no citation.

The cropland mask is free

In Egypt you do not need a land-cover product. Outside the irrigated Nile system there is no water and effectively no rainfall, so seasonal actual evapotranspiration collapses to near zero. A seasonal AETI threshold therefore separates irrigated land from desert almost perfectly — using the same data already being analysed, with no extra dependency, licence or cost. The threshold sits in a genuine gap in the distribution, not in the middle of it.

6. Action and impact

At a \$12.0M budget, the targeted allocation recovers 273 Mm³ against 218 Mm³ for an area-proportional split — 25% more water for the same money, and the same figure on unit cost, so it is not an artefact of spending more.

The prediction we made, tested, and lost

We built this expecting the northern Delta rice belt to lead, reasoning that ponded paddy evaporates all season. The measurement says the opposite: the rice belt holds the most efficient irrigated land in Egypt, and Upper Egypt the least.

Irrigation system	Irrigated area	AETI	Evaporation	E / AETI
Delta — rice belt	1.76 Mha	592 mm	120 mm	0.20
Delta — other	0.61 Mha	500 mm	127 mm	0.26
Nile Valley (Upper Egypt)	1.20 Mha	684 mm	186 mm	0.27

A closed rice canopy shades the water, so ponding does not imply evaporation. Damietta reaches a beneficial fraction of 0.84 and Kafr El Sheikh 0.81 — among the most efficient irrigated landscapes measurable anywhere. Upper Egypt is hotter, carries higher evaporative demand, and grows wider-row crops that leave more bare soil exposed for longer. Without the measurement we would have spent the budget in the wrong half of the country on the strength of a textbook intuition.

How much of that is actually WaPOR?

The question a reviewer will ask, so we answered it first. We re-ran the whole allocation with WaPOR progressively removed.

Information available	Gain vs baseline
Real per-district evaporation — what WaPOR gives you	25%
Evaporation known only as a per-system average — an agronomy textbook	22%
Evaporation spatially flat — no satellite at all	0%

The honest reading. With no spatial information, targeting buys nothing. But most of the value sits in the system-level pattern rather than in per-district detail, so the part attributable to satellite resolution is about **3.7 percentage points**, not 25. That is a weaker claim than we expected to be making, and we are keeping it, because the pattern it points to is exactly the one the satellite told us we had backwards. The information value of a measurement is largest the first time you look.

Pricing the political constraint

Pure cheapest-first funds six governorates out of eighteen. No ministry with an equity mandate signs that, so a recommendation that ignores the constraint is one that gets ignored. Instead of optimising around it, we priced it.

Equity floor	Governorates funded	Water recovered	Cost vs pure targeting
None	7	273 Mm ³	—
25%	18	264 Mm ³	-3.2%
50%	18	253 Mm ³	-7.2%
75%	18	238 Mm ³	-12.9%
100%	18	218 Mm ³	-20.1%

Reserve a quarter of the budget for equity and you fund every governorate for 3.9% of the water. The floor is a live input, so the committee argues with the model in the room and leaves owning the result. The subtheme is explicit that stakeholder engagement happens outside the dataset interface; this is our answer to it.

Recommended allocation

Governorate	Measure	Funded	Mm ³ saved
1. Aswan	Laser land levelling	\$1.19M	29.5
2. Qena	Laser land levelling	\$2.74M	67.8
3. Sohag	Laser land levelling	\$2.17M	50.8
4. Assiut	Laser land levelling	\$2.23M	48.3
5. Beni Suef	Laser land levelling	\$1.97M	41.7
6. Qalyubia	Laser land levelling	\$1.12M	22.9
7. Minya	Laser land levelling	\$0.58M	11.8

Aswan ranks first but carries a low confidence flag and contains Lake Nasser; Qena is the safer headline — highest recoverable volume, medium confidence, no large water body. Units below 5,000 ha are excluded: GAUL's Luxor polygon is 12 km², which is the city rather than the governorate.

The full ranked list, top ten

Governorate	Measure	USD/m ³	Mm ³ available
1. Aswan	Laser land levelling	\$0.040	29.5
2. Qena	Laser land levelling	\$0.040	67.8
3. Sohag	Laser land levelling	\$0.043	50.8
4. Assiut	Laser land levelling	\$0.046	48.3
5. Beni Suef	Laser land levelling	\$0.047	41.7
6. Qalyubia	Laser land levelling	\$0.049	22.9
7. Minya	Laser land levelling	\$0.049	73.1
8. Fayoum	Laser land levelling	\$0.050	46.0
9. Menoufia	Laser land levelling	\$0.051	40.1
10. Sharqia	Laser land levelling	\$0.052	113.1

And because WaPOR observes the same district again next season, the effect of the spending can be measured with the same data that justified it. That closes an accountability loop which currently does not exist.

7. Ecosystem and delivery

- Sits downstream of IPAT, which is already operational inside MWRI — this is an added layer, not a competing tool.
- Delivered as a two-page district brief, the format a ministry memo already uses, exportable in one click.
- Plus a web dashboard where the budget is a slider, so the committee can explore allocations live in the room rather than receiving a fixed recommendation.
- Runs on WaPOR Level 2 (100 m), which is globally available — no dependency on Level 3 coverage, so the method transfers to any WaPOR country.
- Fits the existing annual planning cycle. No new process, no new login, no new institution.
- Fully open: Python pipeline plus a single self-contained HTML file that works offline.

8. What we are not claiming

Three limitations, stated before anyone has to ask.

Field-scale is not basin-scale. Water that evaporates is genuinely lost, but not everything an intervention frees up was evaporating — some was returning to a drain or aquifer and being reused downstream. In a

closed system like the Delta, reuse is high. These figures are for prioritisation, for ranking places against each other. They are not a water account, and the two must not be conflated.

The magnitude is soft; the order is not. Intervention costs and evaporation-reduction fractions are literature-order figures, not measurements, and the size of the prize moves with every one of them. Rather than bury them we made each an editable input. An earlier version of this document claimed the recommendation survived a threefold increase in the cost of alternate wetting and drying; adversarial testing of our own model showed it did not, and we withdrew the claim and corrected the underlying coefficient on physical grounds. The ranking of places is what is robust, because it is driven by observed evaporation, which varies about twofold across the study area.

Alternate wetting and drying saves percolation, not mainly evaporation. Its headline saving in the literature is in water applied. In the closed, high-reuse Delta, percolation is largely recovered downstream, so we deliberately exclude it and count only the evaporation component — removing the free water surface for part of the season. That is why our coefficient is 10%, not the 25–30% usually quoted, and why laser land levelling is the measure the model now selects.

Retrieval is weakest where our signal is. ETLook's evaporation–transpiration partitioning is least constrained over standing water and flooded surfaces — exactly the land surface our result rests on. There is no rebuttal to this. It is a reason for field validation, not for confidence.

The comparator is a neutral prior, not observed practice. We have no citation for how MWRI actually splits this budget, and much of Egypt's on-farm modernisation is delivered through irrigation directorates rather than governorates. We compare against area-proportional allocation because that is what you would do knowing only where the cropland is — not because we claim it is what happens.

9. Next steps

- Replace the demo values with real WaPOR v3 Level 2 rasters for the 2024 summer season.
- Verify the open-water mask on real rasters — Burullus, Manzala, Idku, Maryut and Qarun all sit inside top-ranked governorates. Masking is implemented twice (beneficial fraction in the pipeline, JRC surface water at source) but has not yet run against real data.
- Build the attribution panel: WaPOR evaporation for 2023 and 2024 over funded and unfunded governorates, as a difference-in-differences with the confidence interval shown — including the case where the effect is too small to detect.
- Replace approximate cultivated-area figures with MALR / CAPMAS statistics.
- Swap governorates for MWRI irrigation-directorate boundaries — the single highest-value upgrade, and the one that connects this directly to IPAT's 20 m outputs.
- Put the two-page brief in front of one real MWRI planning officer and record what they say. One quotation from an actual user is worth more in the final pitch than any amount of further modelling.

Sources

IPAT, Irrigation assessment tool for Egypt (MWRI / IWMI / IHE Delft / FAO) — zemamipa.streamlit.app

FAO WaPOR v3, Level 2 100 m dekadal AETI, E, T, I — wapor.apps.fao.org

WaPORIPA, standardised irrigation performance assessment protocol — github.com/wateraccounting/WaPORIPA

FAO WaPOR tools in co-development — fao.org/in-action/remote-sensing-for-water-productivity

Administrative boundaries: Natural Earth admin-1, public domain